

Tools of the Trade: Linux and SQL

Module 1
Introduction to operating systemsModule 2
The Linux operating system

All about Linux

-  Welcome to module 2
Video • 56 sec
-  Introduction to Linux
Video • 3 min
-  Phil: Learn and grow in the cybersecurity field
Video • 2 min
-  Linux architecture
Video • 3 min
-  Linux architecture explained
Reading • 8 min
-  Explore: Components of the Linux OS
Ungraded Plugin • 10 min
-  Test your knowledge: All about Linux
Practice Assignment • Grade: 100%

Linux distributions

-  Linux distributions
Video • 2 min
-  KALI LINUX™
Video • 2 min
-  **More Linux distributions**
Reading • 4 min
-  Package managers for installing applications
Reading • 8 min
-  Resources for completing Linux labs
Reading • 4 min
-  Lab tips and troubleshooting steps
Reading • 4 min
-  Activity: Install software in a Linux distribution
Ungraded App Item • 30 min
-  Optional Exemplar: Install software in a Linux distribution
Ungraded App Item • 10 min
-  Exemplar: Install software in a Linux distribution
Reading • 8 min
-  Test your knowledge: Linux distributions
Practice Assignment • 8 min

The shell

-  Introduction to the shell
Video • 1 min
-  Different types of shells
Reading • 4 min
-  Input and output in the shell
Video • 2 min
-  Activity: Examine input and output in the shell
Ungraded App Item • 30 min
-  Optional Exemplar: Examine input and output in the shell
Ungraded App Item • 10 min
-  Exemplar: Examine input and output in the shell
Reading • 10 min
-  Test your knowledge: The shell
Practice Assignment • 8 min

Review: The Linux operating system

-  Wrap-up
Video • 51 sec

More Linux distributions

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Previously, you were introduced to the different distributions of Linux. This included KALI LINUX™. (KALI LINUX™ is a trademark of OffSec.) In addition to KALI LINUX™, there are multiple other Linux distributions that security analysts should be familiar with. In this reading, you'll learn about additional Linux distributions.

KALI LINUX™

KALI LINUX™ is an open-source distribution of Linux that is widely used in the security industry. This is because KALI LINUX™, which is Debian-based, is pre-installed with many useful tools for penetration testing and digital forensics. A **penetration test** is a simulated attack that helps identify vulnerabilities in systems, networks, websites, applications, and processes. **Digital forensics** is the practice of collecting and analyzing data to determine what has happened after an attack. These are key activities in the security industry.

However, KALI LINUX™ is not the only Linux distribution that is used in cybersecurity.

Ubuntu

Ubuntu is an open-source, user-friendly distribution that is widely used in security and other industries. It has both a command-line interface (CLI) and a graphical user interface (GUI). Ubuntu is also Debian-derived and includes common applications by default. Users can also download many more applications from a package manager, including security-focused tools. Because of its wide use, Ubuntu has an especially large number of community resources to support users.

Ubuntu is also widely used for cloud computing. As organizations migrate to cloud servers, cybersecurity work may more regularly involve Ubuntu derivatives.

Parrot

Parrot is an open-source distribution that is commonly used for security. Similar to KALI LINUX™, Parrot comes with pre-installed tools related to penetration testing and digital forensics. Like both KALI LINUX™ and Ubuntu, it is based on Debian.

Parrot is also considered to be a user-friendly Linux distribution. This is because it has a GUI that many find easy to navigate. This is in addition to Parrot's CLI.

Red Hat® Enterprise Linux®

Red Hat Enterprise Linux is a subscription-based distribution of Linux built for enterprise use. Red Hat is not free, which is a major difference from the previously mentioned distributions. Because it's built and supported for enterprise use, Red Hat also offers a dedicated support team for customers to call about issues.

AlmaLinux

AlmaLinux is a community-driven Linux distribution that was created as a stable replacement for CentOS. CentOS was an open-source distribution that is closely related to Red Hat, and its final stable release, CentOS 8, was in December 2021. CentOS used source code published by Red Hat to provide a similar platform. AlmaLinux is designed to be a drop-in replacement for CentOS 8. This ensures that applications and configurations that worked on CentOS will continue to function on AlmaLinux.

Key takeaways

KALI LINUX™, Ubuntu, Parrot, Red Hat, and CentOS are all widely used Linux distributions. It's important for security analysts to be aware of these distributions that they might encounter in their career.

Mark as completed

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Venkatesh

Give me a summary



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This reading explores various Linux distributions that are important for security analysts to know, beyond just KALI LINUX™.

Specialized Security Distributions

- **KALI LINUX™**: A Debian-based, open-source distribution pre-installed with tools for penetration testing and digital forensics.
- **Parrot**: Another Debian-based, open-source distribution with pre-installed tools for penetration testing and digital forensics, known for its user-friendly GUI.

General Purpose and Enterprise Distributions

- **Ubuntu**: A user-friendly, open-source, Debian-derived distribution with a large community, widely used in security and cloud computing.
- **Red Hat® Enterprise Linux®**: A subscription-based, enterprise-grade Linux distribution that offers dedicated support.

Community-Driven Alternatives

- **AlmaLinux**: A community-driven distribution designed as a stable replacement for CentOS, ensuring compatibility with applications and configurations that ran on CentOS 8.

And if you want to continue exploring this topic, try one of these follow-up questions:

What is the main difference between KALI LINUX™ and Red Hat Enterprise Linux?

How can you use Ubuntu for cloud computing security tasks?

How would switching from CentOS to AlmaLinux impact system compatibility and security?



Ask me anything



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